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WIFI CAMERA SMART ROBOT RESEARCH

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Background of the Study

The development of WiFi Camera Smart Robots has been a significant advancement in home security systems, particularly in the context of Internet of Things (IoT) technologies. These devices offer remote surveillance through Wi-Fi wireless data transfer, allowing users to monitor their homes in real time, receive alerts, and record footage. This technology is especially beneficial for individuals with mobility impairments who are unable to move freely around their homes to conduct surveillance. By leveraging home Wi-Fi networks, WiFi Camera Smart Robots provide a sense of security and independence to individuals who may otherwise feel vulnerable.

Existing research has explored the general functionality of IoT devices, including smart robots used for surveillance. Numerous studies highlight the benefits of wireless camera systems over traditional wired security systems, focusing on convenience, ease of installation, and flexibility. In addition, some research has addressed the limitations of such systems, including issues like signal strength and the potential for interruptions, especially in areas with less reliable Wi-Fi infrastructure.

However, a gap remains in the literature regarding the specific needs of individuals with mobility challenges in relation to smart surveillance robots. While general studies have focused on the use of these devices in the broader population, little research has been dedicated to understanding how these technologies can best support people with disabilities. Furthermore, there is limited exploration into how factors such as signal interruption, Wi-Fi strength, and environmental conditions can impact the reliability and effectiveness of Wi-Fi camera systems, especially in regions where high-tech solutions like Starlink are not readily available or affordable.

What is Missing or Open Questions:

Although there is existing research on IoT surveillance devices, several key questions remain unanswered:

> **Signal Interruption**: While modern Wi-Fi systems are improving in terms of reliability, interruptions still pose a potential risk. What impact do signal interruptions have on the effectiveness of WiFi Camera Smart Robots, and what measures can be taken to mitigate this risk, particularly for people with mobility challenges?

Comparison with Traditional Systems: How do Wi-Fi camera smart robots compare to traditional wired security cameras in terms of performance, security, and ease of use, especially for users with limited mobility?

Affordability and Accessibility: Not all users have access to advanced technologies like Starlink, which offer robust Wi-Fi connections. How do variations in Wi-Fi strength affect the performance of WiFi Camera Smart Robots, and what alternatives can be offered to users in areas with weaker connections?

These gaps in the literature emphasize the need for more research to better understand the specific needs of people with disabilities and the practical challenges of using Wi-Fi-based surveillance robots in different contexts.

Rationale for the Study:

Given the increasing reliance on Wi-Fi-based security systems, it is crucial to address the gaps in existing research regarding the use of WiFi Camera Smart Robots for individuals with mobility impairments. This study aims to explore how these devices can provide practical solutions to enhance safety and independence for people who cannot easily check their homes due to physical limitations. By investigating the effects of signal interruptions, comparing Wi-Fi cameras with traditional wired systems, and evaluating how factors like Wi-Fi strength impact functionality, this research will provide valuable insights into optimizing the use of smart surveillance technologies for people with disabilities.

Moreover, this study will contribute to the broader understanding of IoT applications in home security, particularly for users in regions with varying levels of access to high-speed internet services.

Hypothesis:

WiFi Camera Smart Robots can offer an effective and reliable solution for home surveillance, particularly for individuals with disabilities. Despite challenges such as signal interruptions, advancements in Wi-Fi technology can mitigate these issues, making WiFi camera systems a viable alternative to traditional wired cameras. However, the success of these devices may depend on the quality of the Wi-Fi connection and the specific needs of the user.

Objectives:

General Objectives:

To assess WiFi Camera Smart Robots' overall efficacy, usability, and impact in improving home security and helping people with mobility impairments, especially in settings with inconsistent Wi-Fi.

Specific Objectives:

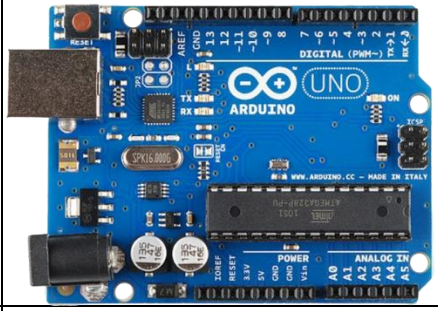
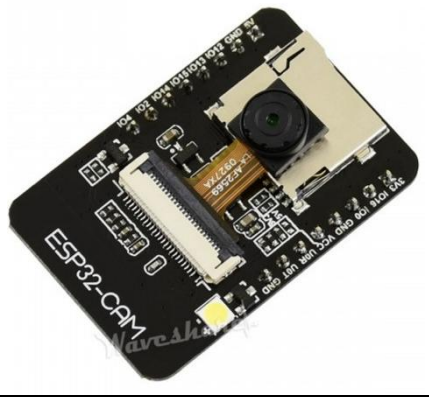
- To explore the components and functionalities of WiFi Camera Smart Robots and how they support people with mobility impairments.
- To assess the reliability and effectiveness of WiFi Camera Smart Robots in environments with varying Wi-Fi strength, addressing issues such as signal interruptions.
- To compare WiFi Camera Smart Robots to traditional wired security cameras in terms of usability, security features, and overall user satisfaction, especially for people with disabilities.
- To identify and evaluate strategies to improve the performance of WiFi Camera Smart Robots in homes with weak Wi-Fi connections.

A. INTRODUCTIONG OF THE PROJECT OR STUDY

The methodology of our study explains how we created and understood the WiFi Camera Smart Robot, and how each part works together to make the project useful especially for people who cannot walk properly and for those persons who want to monitor and check the surroundings of their houses without exerting an effort. In this part, we discuss the parts of the robot, how they are used, how we connected

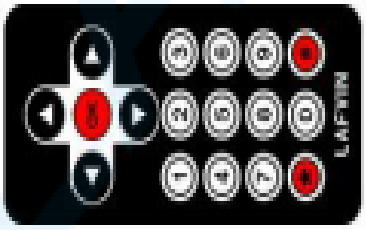
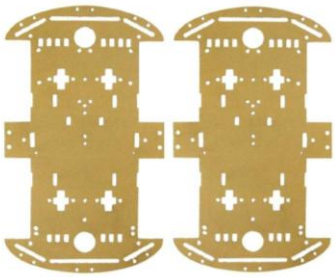
everything, and how we tested the project so we can see if it really works in real situations just like what we talked about in the introduction.

B. DISCUSSION OF THE PARTS

Component Name	Function	Discription
Arduino Uno R3		The main controller board that runs the robot's program and connects all sensors and motors.
L298N Motor Driver Module		Controls the speed and direction of the DC motors using signals from the Arduino.
ESP32-CAM Module		A small camera board that can take photos, stream video, and send data through WiFi.

ESP32-CAM-MB		A USB programming board used to upload code easily to the ESP32-CAM.
KY-033 Line Tracking Sensor		Detects black and white lines on the ground for line-following functions.
External Antenna		Boosts the WiFi signal of the ESP32-CAM for better range and stable video.

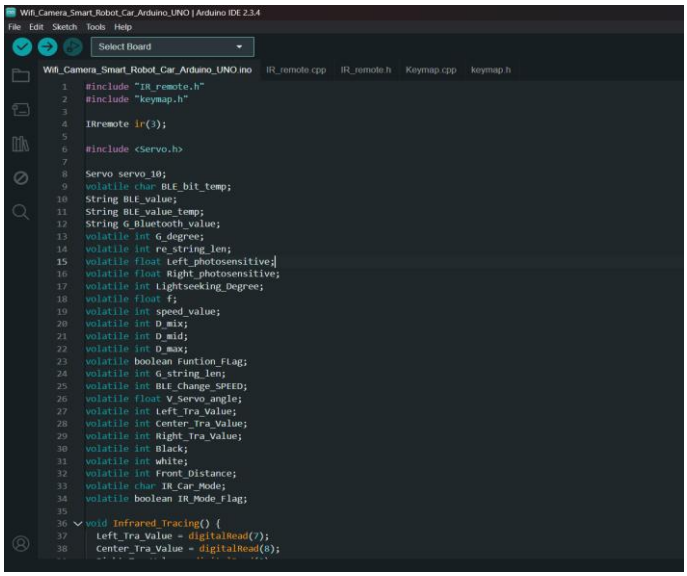
DC Car Wheels		The wheels that allow the robot to move forward, backward, and turn.
Ultrasonic Sensor HC-SR04		Measures distance using sound waves, helping the robot avoid obstacles.
FS90R micro sirvo		A continuous-rotation servo used for precise turning or controlling parts like a rotating sensor mount.

Dc geared motors		Motors with Built-in gears for stronger torque to drive the robot’s wheels smoothly
Remove control		Allows the user to manually control the robot wirelessly
USB 2.0 Type A to Type B cable		Used to connect the Arduino Uno R3 to a computer for uploading code and power
Smart Robot Car Chassis Kit		The physical frame of the robot where all components such as motors, sensors, and wheels are mounted

ESP32 camera case		Protects the ESP32-cam and keeps the camera stable during robot movement
Battery holder		Holds and connects the batteries that power the robot’s components
Jumper wires		Flexible wires used to connect sensors, motors, and modules to Arduino.
Micro USB		Used to power or program modules like the ESP32-cam-mb

GPTOP lithium battery 186503.7v		Rechargeable battery that provides power to the robot's motors and electronics.

Program Code and Functionality



The Arduino program (Wifi_Camera_Smart_Robot_Car_Arduino_UNO.ino) controls a robot car by processing inputs from Bluetooth/Serial, an IR Remote, and sensors for line tracing and distance measurement. Its main functionality is executed in a continuous loop() that first handles Bluetooth commands followed by IR remote controls. Movement is managed through functions such as Move_Forward, Rotate_Left, and STOP, utilizing digital and PWM pins for motor control. The robot operates in various modes: Manual Control, Infrared Tracing, Ultrasonic Avoidance, and Ultrasonic Follow. Continuous modes rely on sensor feedback for movement decisions, and the IR remote can adjust a servo motor's angle. Commands received over serial are parsed using a specific protocol starting with '%' and ending with '#'.

```
#include "IR_remote.h"
#include "Keymap.h"
// Provides ISR
#ifndef __AVR_ATmega32U4__
#include <avr/interrupt.h>

volatile irparams_t irparams;
bool MATCH(uint8_t measured_ticks, uint8_t desired_us)
{
  // Serial.print(measured_ticks);Serial.print(",");Serial.println(desired_us);
  return(measured_ticks >= desired_us - (desired_us>>2)-1 && measured_ticks <= desired_us + (desired_us>>2)+1);
}

ISR(TIMER_INTR_NAME)
{
  // Serial.println("ISR");
  //Serial.println(millis());
  uint8_t irdata = (uint8_t)digitalRead(irparams.recvpin);
  // uint32_t new_time = micros();
  // uint8_t timer = (new_time - irparams.lastTime)>>6;
  irparams.timer++; // One more 50us tick
  if (irparams.rawlen >= RAWBUF)
  {
    // Buffer overflow
    irparams.rcvstate = STATE_STOP;
  }
  switch(irparams.rcvstate)
  {
    case STATE_IDLE: // In the middle of a gap
      if (irdata == MARK)
      {
        irparams.rawlen = 0;
        irparams.timer = 0;
        irparams.rcvstate = STATE_MARK;
      }
      break;
    case STATE_MARK: // timing MARK
```

```
#ifndef IRremote_h
#define IRremote_h

/* Includes ------------------------------------------------------------------*/
#include <stdint.h>
#include <stdbool.h>
#include <Arduino.h>

#ifdef ME_PORT_DEFINED
#endif // ME_PORT_DEFINED
#ifndef __AVR_ATmega32U4__
#define MARK  0
#define SPACE 1
#define NEC_BITS 32
#define USECPERTICK 50  // microseconds per clock interrupt tick
#define RAWBUF 80 // Length of raw duration buffer

typedef enum {ERROR = 0, SUCCESS = !ERROR} ErrorStatus;

#define NEC_HDR_MARK  9000
#define NEC_HDR_SPACE 4500
#define NEC_BIT_MARK  560
#define NEC_ONE_SPACE 1600
#define NEC_ZERO_SPACE  560
#define NEC_RPT_SPACE 2250
#define NEC_RPT_PERIOD  110000


#define _GAP 5000 // Minimum map between transmissions

// receiver states
#define STATE_IDLE     2
#define STATE_MARK     3
#define STATE_SPACE    4
#define STATE_STOP     5
```

The IR remote library is designed for receiving and decoding **NEC infrared signals** on an Arduino board (specifically excluding the ATmega32U4). It uses a **Timer2 interrupt** running at 50 microseconds per tick (USECPERTICK) to precisely measure the duration of IR marks (ON) and spaces (OFF), storing these raw intervals in a buffer (rawbuf). The interrupt service routine (ISR) manages a state machine (IDLE, MARK, SPACE, STOP) to capture the incoming signal, and once a long space/gap is detected, the signal is ready for decoding.
The primary function, decodeNEC(), analyzes the collected raw timing data against expected NEC protocol values (Header Mark, Header Space, Bit Mark, One Space, Zero Space) to extract a 32-bit data value. The library also includes functions for transmitting IR signals, such as sendNEC() and sendRaw(), which utilize the Arduino's PWM capabilities to generate the 38 kHz carrier frequency. Helper functions like getKeyMap() and getIrKey() translate the raw decoded IR code into standardized key names or indices defined in the accompanying Keymap.h file.

```
#include "Keymap.h"
ST_KEY_MAP normal_ir_keymap[18] = {
  {"1", 0x16},
    {"2", 0x19},
    {"3", 0x0d},
    {"4", 0x0c},
    {"5", 0x18},
    {"6", 0x5e},
    {"7", 0x08},
    {"8", 0x1c},
    {"9", 0x5A},
  {"0", 0x52},
    {"*", 0x42},
    {"#", 0x4A},
  {"up", 0x46},
  {"down", 0x15},
  {"ok", 0x40},
    {"left", 0x44},
  {"right", 0x43},
};//logo LA
```

```
#ifndef _KEYMAY_H_
#define _KEYMAY_H_
#include <Arduino.h>
#define KEY_MAX 21
typedef struct
{
    String keyname;
    byte keycode;
}ST_KEY_MAP;

#define IR_TYPE_NORMAL 1
#define IR_TYPE_EM 2

typedef enum {
    IR_KEYCODE_1 = 0,
    IR_KEYCODE_2,
    IR_KEYCODE_3,
    IR_KEYCODE_4,
    IR_KEYCODE_5,
    IR_KEYCODE_6,
    IR_KEYCODE_7,
    IR_KEYCODE_8,
    IR_KEYCODE_9,
    IR_KEYCODE_0,
    IR_KEYCODE_STAR,     // *
    IR_KEYCODE_POUND,    // #
    IR_KEYCODE_UP,
    IR_KEYCODE_DOWN,
    IR_KEYCODE_OK,
    IR_KEYCODE_LEFT,
    IR_KEYCODE_RIGHT,
}E_NORMAL_IR_KEYCODE;

typedef enum {
    EM_IR_KEYCODE_A = 0,
    EM_IR_KEYCODE_B,
    EM_IR_KEYCODE_C,
    EM_IR_KEYCODE_D,
```

In order to read decoded infrared signals, the key mapping files provide a link between physical infrared remote buttons and their digital keycodes. Two enumeration types, E_NORMAL_IR_KEYCODE and E_EM_IR_KEYCODE, are defined in the header file keymap.h. These types provide understandable names (such IR_KEYCODE_UP) for key indices on different remote types. The key name and its byte keycode are stored in the ST_KEY_MAP structure. The normal_ir_keymap array is implemented by Keymap.cpp, which links string names to their corresponding 8-bit hexadecimal keycodes. This allows the IRremote library's functions to convert raw NEC signals into recognized commands for robot motion or servo control.

C. DISCUSS THE SCHEMATIC DIAGRAM

1. Power & Motor Control

The code uses four pins for the two DC drive motors, which implies control via an H-bridge Motor Driver for direction and speed.

Component	Arduino UNO Pin	Function
Left Motor (Direction)	Digital Pin **4**	High/Low for motor direction (Input 1 on driver)
Left Motor (Speed/PWM)	Digital Pin **5**	PWM output for speed (Enable A on driver)
Right Motor (Direction)	Digital Pin **2**	High/Low for motor direction (Input 3 on driver)
Right Motor (Speed/PWM)	Digital Pin **6**	PWM output for speed (Enable B on driver)
Motor Driver	**VIN/GND**	Connected to a separate 5-12V battery pack.

2. Sensors

The robot uses an Ultrasonic distance sensor and three Infrared line-tracing sensors.

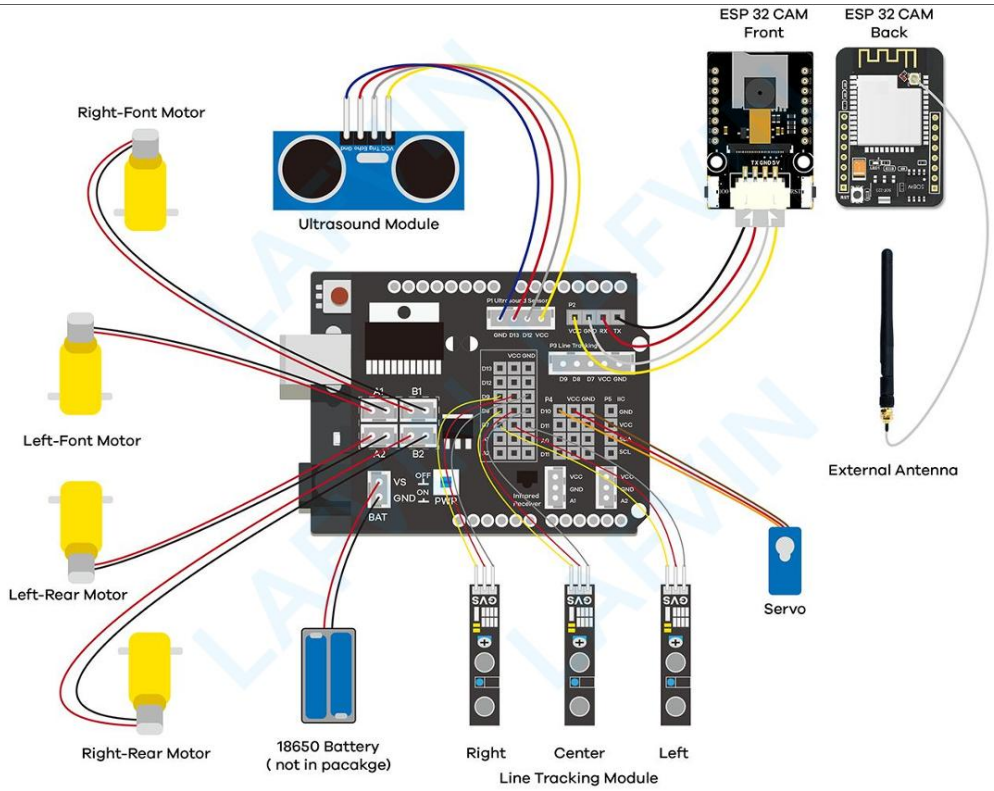
Component	Arduino UNO Pin	Function
IR Receiver	**Digital Pin 3**	**Input for decoding remote control signals.**

Component	Arduino UNO Pin	Function
Ultrasonic Trigger (Trig)	Digital Pin 12	Output pulse to initiate distance measurement.
Ultrasonic Echo (Echo)	Digital Pin 13	Input to measure the return pulse duration.
Left Tracing Sensor	Digital Pin 7	Input to read line reflectivity (digital).
Center Tracing Sensor	Digital Pin 8	Input to read line reflectivity (digital).
Right Tracing Sensor	Digital Pin 9	Input to read line reflectivity (digital).

3. **Communication & Actuators**

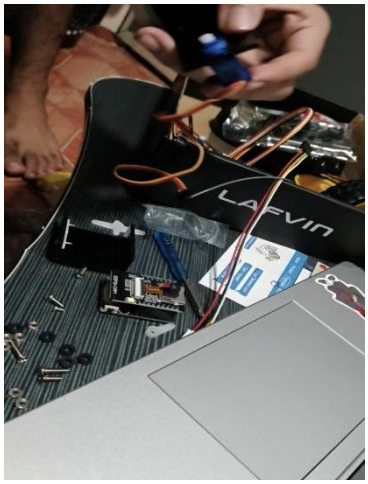
   The communication uses the primary hardware Serial port, which is often connected to a Bluetooth or WiFi module. The servo is likely used to pan a sensor or the camera.

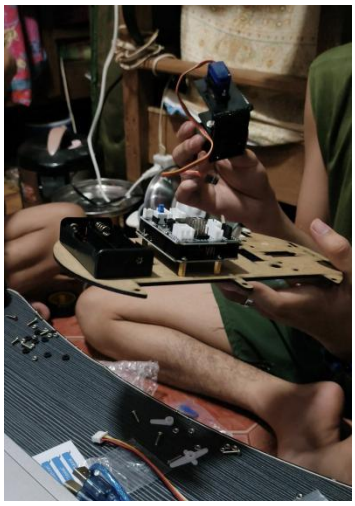
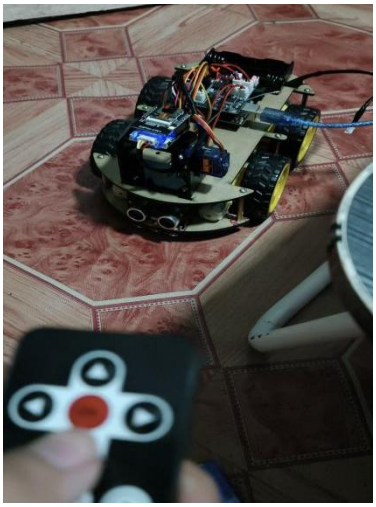
Component	Arduino UNO Pin	Function
Bluetooth/WiFi Module (RX)	Digital Pin 0 (RX)	Receives data from the module.
Bluetooth/WiFi Module (TX)	Digital Pin 1 (TX)	Sends data to the module.
Servo Motor	Digital Pin 10 (PWM)	Controls the servo angle (servo_10.write).



This wiring diagram illustrates a highly functional and versatile **smart robot car** architecture centered on the **Arduino Uno R3** and a **L298N Motor Driver Module**. The system efficiently integrates multiple control and sensing capabilities, using the **L298N Motor Driver Module** to drive the **DC geared motors** for movement and control the **micro servo**. Essential environmental awareness is provided by the **Ultrasonic Sensor** (for distance) and the **Line Tracking Sensor** (for navigation), allowing for autonomous behaviors like obstacle avoidance and line-following. Furthermore, the inclusion of the **ESP32-CAM module** and its associated board add's advanced features, enabling **Wi-Fi connectivity** for remote control and video streaming. Ultimately, the design creates a powerful, multi-modal educational platform capable of executing complex tasks by combining traditional microcontroller control with modern IoT capabilities.

D. PICTURE OF THE WORKING AND FINAL PROJECT



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